Modeling the Impact of VANET-Enabled Traffic Lights Control on the Response Time of Emergency Vehicles in Realistic Large-Scale Urban Area

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Abstract— **In the last decade, intelligent transportation systems (ITS) have progressed at a rapid rate, which aim to improve transportation activities in terms of safety and efficiency. Car-to-car and car-to-infrastructure communications are important components of the ITS architecture. Communication between cars and traffic lights is one of the important V2I applications which helps to have dynamic and automatic traffic lights that can create several benefits such as minimizing the traffic jam, reducing fuel consumption and emissions, etc. This paper deals with decreasing the response time of the emergency cars by changing the traffic lights status with employing the communication technologies. The contribution of this paper is twofold: First, the effect of the changing traffic lights status to green for emergency cars is investigated by using traffic simulator (SUMO). Second, this paper uses OMNET++ (Network Simulator) in order to simulate the mentioned scenario as a VANET (with 802.11p standard) by using Veins framework to run SUMO and OMNET++ in parallel. This study has developed the Veins framework by adding a new module to OMNET++ to consider the traffic lights which are simulated in SUMO. Moreover this study has developed a new program written in Python which is connected to SUMO and controls the traffic simulation. This program uses SUMO to simulate a microscopic traffic (by considering every single vehicle movements) and also a city with intelligent traffic lights. Additionally, several statistics about traffic simulation is created for each car such as traveling time, waiting time, emissions, fuel consumption; or complete amount of car emissions in the street during the simulation, fuel consumption, number of vehicles and so on, for each street. This paper uses Manhattan realistic map to describe the mentioned program, then uses realistic map and realistic traffic demand of Cologne, Germany, to obtain a realistic and reliable result.**

Keywords-component; intelligent transportation systems (ITSs); vehicle-to-infrastructure communications(V2I); intelligent traffic lights; VANET; OMNET++; SUMO; Veins; 802.11p; Beaconing; Emergency Vehicles response time.

I. INTRODUCTION

Recently increasing the number of cars on city roads has created many problems, such as traffic congestion, the huge number of people who get killed in car accidents, fuel consumption, emissions, etc. For example, according to the National Highway Traffic Safety Administration (NHTSA) there are about 43,000 people who are killed in fatal car accidents each year in the United States [1]. One of the most important problems which is happening because of traffic congestion is the increasing emergency vehicle response time for an incident and this issue should be considered. (An Emergency vehicle is the general term for ambulances, fire trucks, police cars, etc.). For example, to illustrate the importance of this issue, delivery efficiency of emergency medical services (EMS) is critical in reducing mortality and disability rates. Several studies have been done to show the important relationship between response time and mortality rate [2]-[3]. An EMS response time can be defined as the interval from the time that call was received by the EMS provider to the arrival of the ambulance to the emergency scene[4]-[5]. Another example is that the mortality in the car accidents can be categorized as follows: A) Death occurs in the first few seconds or minutes after an accident (about 10% of all deaths). B) The first hour after the accident, so-called Golden Hour, has the highest mortality (75% of all deaths). C) Death occurs in the days or weeks after the accident [6].

Thus, it is clear that decreasing the response time for any public emergency car is one of the most important problems in new urban transportation systems that could save numerous people lives. To solve this problem the easiest way is to build new road infrastructures or increase the number of emergency cars, unless resources such as time, money, and space are limited. Thus, there is a need to manage the existing road infrastructures effectively and safely. Wireless communication technology is one of the useful solutions in order to solve the traffic problems. Car to Car Communication (C2CC) or Vehicle to Vehicle (V2V) Communication and Vehicle to Infrastructure (V2I) Communication based on Wireless technology is designed to help cars to “talk” to each other and also to Road Side Units (RSU). Due to high expenses and dangers of testing new technologies for transportation in the real world, simulation plays an important role to find out the beneficial and effective technologies before implementation. In the last decade, several traffic simulators are developed so that they can simulate real traffic of cars easily, but simulation of emergency cars traveling in large real areas is difficult yet. This study has developed a program which is connected to a microscopic traffic simulator, SUMO[7], and controls the cars and traffic lights. The main goal of this program is changing the traffic lights status automatically based on distance from a

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specific emergency vehicle. Also several statistics of traffic simulation is created. This program simulates traveling of emergency cars in a real city with real traffic in a straightforward manner. This paper focuses on decreasing the response time of public emergency cars by using communication between an emergency car and traffic lights. This can be done by changing the status of traffic lights in the emergency car's route.

The remainder of this paper is organized as follows: Section II briefly explains the traffic simulator and describes the program that is developed in this study by using Manhattan map. Then Section III presents the scenario of the realistic map and realistic traffic which is simulated by SUMO and the mentioned program and the result for this section are discussed. Section IV explains VANET simulation by using OMNET++, SUMO and Veins framework and at the end of the section, the VANET simulation results are discussed. This paper is concluded in Section V and the future work is proposed.

II. Traffic Simulation

A. SUMO

Vehicular Mobility Simulations are usually divided into two main types: microscopic and macroscopic. Macroscopic only focuses on flow mobility of cars not each car individually. In Macroscopic simulation, generations of vehicular traffic such as traffic density or traffic flows are defined. However in the microscopic approach the movement of each individual vehicle and the vehicle behavior is important [8]. Microscopic model is used in V2I and V2V simulations. Researchers use various simulators like SUMO, VanetMobiSim, CORSIM, CityMob, VISSIM, STRAW, PARAMICS, FreeSim and Netstream. Based on being open-sourced, simple real map importing, several tools for connection to network simulators like OMNET++ or NS-2, supporting different real traffic demand and realistic traffic simulation, SUMO (Simulation of Urban MObility) is selected as mobility generator in this study.

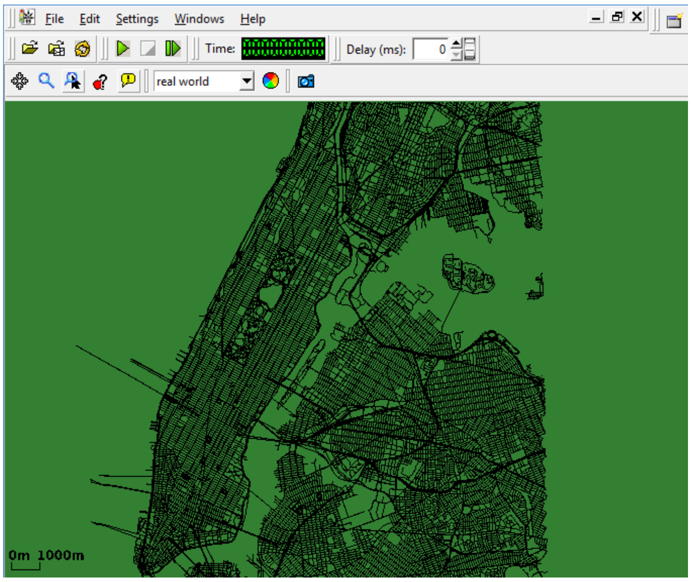

Figure 1. Manhattan map in SUMO

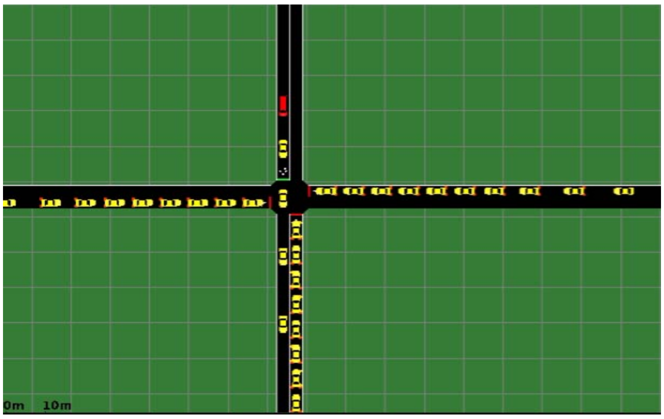

Figure 2. Emergency Car in Crossroad

Simulation in SUMO requires two types of input, Street Network (Map) and Traffic demand. In this paper, street network of the cities are imported from OpenStreetMap (OSM) database[9]. The traffic demand could be imported from different sources like using flow definitions, Randomization, OD-matrices, etc. In addition SUMO is able to connect to other software via TraCI. Traffic Control Interface (TraCI) gives the possibility to control a running road traffic simulation. Therefore, a TCP-based client/server architecture is established where SUMO acts as a server and the "controller" is the client. This paper has developed a program written in Python which is connected to SUMO and controls the simulation.

B. Developed Program

The mentioned program is able to simulate a city with intelligent traffic lights which means it is capable of changing traffic lights status according to the required demand. As an example changing the traffic lights status based on distance from an emergency car in Manhattan is illustrated. Figure.1 shows the Manhattan map in SUMO. In addition 100,000 cars have been added to the map.

The program is started by requesting the name of the emergency car and the route. Hence, the specified path from origin to destination is selected in Manhattan map. Then the emergency car which has higher priority than the other cars is added to the simulation and the simulation is started. All traffic lights in the route have been considered. At each step of the simulation, the distance between the emergency car and the traffic lights is calculated. If the distance is less than the specified value (which is defined at the beginning of the simulation), the traffic light status changes to green for the emergency vehicle and red for the other cars (Fig. 2). After the emergency vehicle passes the traffic light (with a distance of 20 meters) the traffic light status returns to its original state.

The program has generated several statistics which Table 1 shows the important ones. In addition the status of traffic lights at every instant of time could be observed. As clarified before, the main goal of this paper is to decrease the response time of emergency vehicles by changing the status of traffic lights to decrease waiting time in junctions. In order to achieve a realistic and reliable result, real map and real traffic of cars are required. The next section describes the real map and traffic with simulation scenario to get a desirable result.

TABLE 1. SIMULATION STATISTICS

Every Single Vehicle (Also Emergency Vehicle)	**Every Street**
Speed, acceleration, position at every instant.	Maximum and average speed of cars in the street
Length of the vehicle route and Traveling time.	Average traveling time on the street
Fuel Consumption	Complete amount of fuel consumption
Emitted gas (CO, CO2 ,HC, PMx, NOx)	Complete amount of gas emitted by vehicles on the lane (CO, CO2, HC, PMx, NOx)
All traveled streets and Junctions	Number of cars on the street
Emitted Noise	Emitted noise by cars on the street

III. TRAFFIC SIMULATION WITH SUMO

A. Real Map and Realistic Traffic

For road network and traffic demand, a data set from [10] has been used. The realistic traffic demand for city of Cologne, in Germany with a realistic map for this city have been presented in [10]. Travel of the cars dataset is mainly based on data made by TAPAS-Cologne project. TAPASCologne, provided by the Institute of Transportation Systems at the German Aerospace Center (ITS-DLR), determines realistic car traffic in the city of Cologne. Also the street network of the city is imported from OpenStreetMap (OSM) database[9]. This data covers an area of approximately 400 km^2 around Cologne for a period of 24 hours, comprising more than 700.000 individual car trips. These two dataset has been exported to SUMO. Fig. 3 shows the Cologne map in SUMO.

B. Simulation Scenario

There is no official standard for response time of Public emergency services; however response time standards frequently do exist in the form of contractual obligations between communities and emergency service provider organizations. For example in Montreal, QC, Canada, 90% of ambulance requests should be served within 7 minutes [11], or based on standard rights in the urban areas in United States, 95% of requests should be served within 10 min, whereas, in rural areas, they should be served within 30 min[12]. In order to achieve desired response time, several components should be considered such as location of emergency vehicles station, number of vehicles in the station, population around the station, maximum coverage area of the station, etc. To investigate the response time, this paper assumes that the maximum travel distance for emergency vehicles is 5 km with average speed of 60 km/h (based on data from [4]). Then simulation scenario is defined by using the Cologne map and traffic, the developed program, and SUMO as follows:

1) Initially, the map of Cologne is divided into several zones, and realistic traffic of cars between 6 am till 8 am is considered.

2) After that, 20 different zones with different traffic density are selected based on the real traffic status at 7:00 am (Fig. 4). Then, an emergency car is added at each zone and an identical traveling distance (5km) is defined for all the cars.

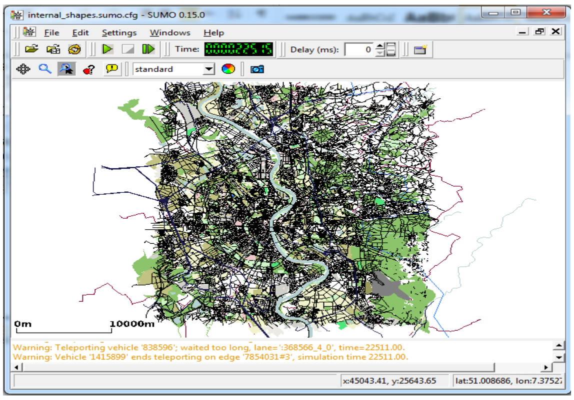

Figure 3. Cologne map in SUMO

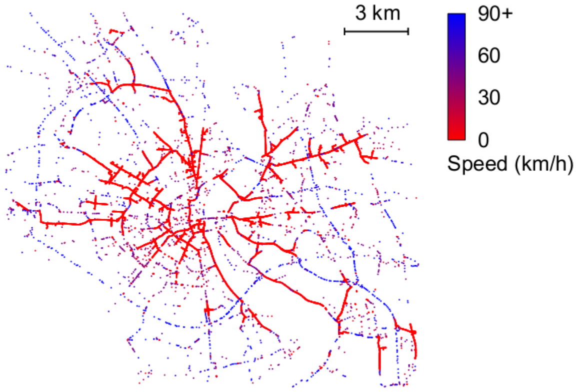

Figure 4. Original TAPASCologne dataset. Snapshot of the traffic status at 7:00 am, in a 400 km^2 region centered on the city of Cologne. Blue vehicles are moving, whereas bright red ones are still[13].

3) To find out the effect of changing traffic lights status in emergency vehicles response time based on distance between emergency car and traffic lights, five different simulations are done as follows:

a. Firstly, only 20 emergency cars travel in the Cologne map without any other car or any traffic light. Then traveling time (response time) and traveling distance for 20 emergency cars are obtained.

b. The second simulation is done with the realistic traffic of city of Cologne and also with 20 emergency cars. This means more than 250.000 individual cars moving in cologne in this two hours. Also in this simulation the traveling time for 20 emergency cars are obtained.

c. In the third simulation the status of some of traffic lights are changed as follows: the route of an emergency car is considered and if the distance between the emergency car and traffic light is less than 50 meters, the traffic light status changes to green for emergency car and red for others. After the emergency car passes the traffic lights (with the distance of 20 meters) the traffic status returns to their original state. This changing status of traffic lights is done for all 20 emergency cars and the traveling times are obtained.

d. The fourth simulation is done similar to the third one with using the 300 meters distance for changing traffic light status and the traveling time is obtained.

e. Last simulation is done like previous one with using a 5 Km distance between traffic light and an emergency car to change the status of the traffic lights.

C. *Traffic Simulation Results*

Simulations are done with the explained scenario. All data are collected and ordered based on traveling time for emergency cars (Response Time) from low to high in Fig. 5. It is obvious that changing the traffic lights status to green for emergency cars decreases the response time. In order to obtain a reasonable distance for changing traffic light status another factor is required. This paper selects cars delay time which occurs because of changing the traffic light status. It means that the route of each emergency car is considered and delay time for other cars that have stopped at traffic light due to the mentioned changing traffic light status is considered and maximum delay is obtained. Fig. 6 illustrates the maximum delay time for 20 cars that stopped in each emergency cars route. As it can be seen in Fig.5 changing the traffic light status for each route has a different effect. For example, for changing distance equal to 50 meters, in low traffic area (car's number 1-7) the average decrease of the response time is 36.39%, for medium traffic area (car's No. 7-14) is 16.66% and for high traffic area (car's No. 14-20) is 6.93%.

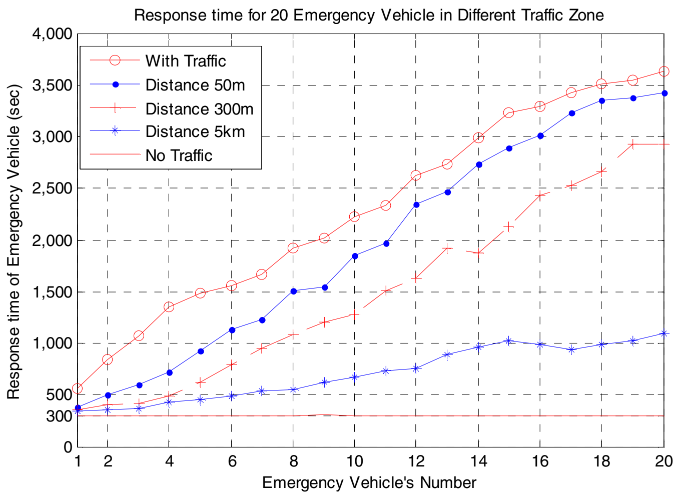

Figure 5. Response Time for Emergency Cars with changing Traffic Lights Status based on distance between cars and traffic lights.

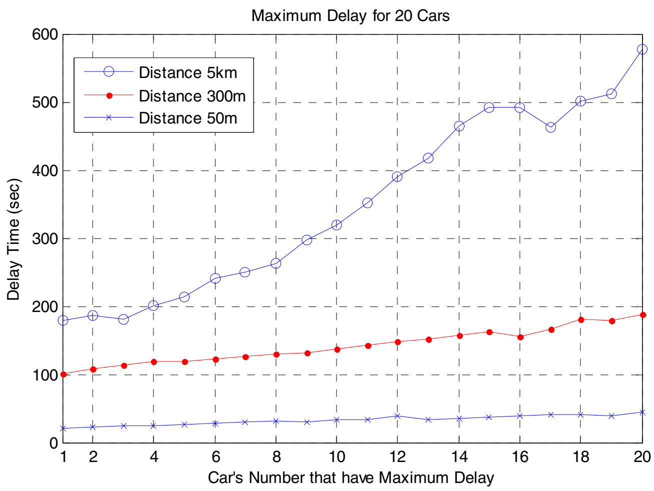

Figure 6. Maximum Delay for 20 cars that have stopped in traffic lights because of Emergency Vehicles.

IV. VANET SIMULATION

A. *VANET*

The basic concept of the VANET (Vehicular Ad-hoc Network) is straightforward and simple: create a widespread and cheap wireless technology to connect vehicles to each other and road side units (RSU) for sending and receiving the information. The important features of VANETs includes these items: Cars and RSU are considered as nodes in VANET, The nodes can move very fast, and the network is highly dynamic which means that topology of the network is changing continuously with changing the position of the nodes and density[14]. Simulation of the VANET protocols, such as wireless communication, multi hop routing, etc, is typically performed in network simulation environments such as OMNET++ or NS-2(3). However, network simulators cannot simulate the mobility of cars sufficiently. Duo to this fact, to have a realistic and authentic result, the mobility of cars should be simulated by a trusted mobility simulator such as SUMO. In order to run the network and traffic simulator in parallel, another software is required. There are several interfaces to couple the network simulators and traffic simulator. [15] has presented the MOVE software that is based on NS-2 and SUMO. [16] has worked on TraNS which integrates SUMO and NS-2. VanetMobiSim simulator which is written in Java has been studied in [17]. In [18]real maps are used to create mobility traces. The integration of SUMO and NS-2 is done in [19]. Recently a simulation framework with integration between NS-3 and the microscopic traffic simulator DIVERT[20] is presented in[21] and then is extended in [22]. One of The drawbacks of most integrated simulators is that the network simulator cannot influence the cars behavior in the mobility simulator.This paper selects Veins[23] framework which has coupled OMNET++ and SUMO due to these most important features: Online re-configuration and re-routing of cars in reaction to network simulator, Fully-detailed models of IEEE 802.11p and IEEE 1609.4 DSRC/WAVE network layers, Supporting the realistic map and realistic traffic. Veins enables running of two simulators in parallel, connected via a TCP socket. Veins framework is developed based on MiXiM. OMNeT++ provides a powerful networking simulation tool but it has deficiency in modelling of wireless communication. In [24] MiXiM, a framework for simulating wireless channels has been presented which provides detailed models of wireless channels, connectivity, mobility and MAC layer protocols for OMNeT++.

B. *Beaconing*

The C2CC concept relies on continuous broadcast of information by all the cars and RSUs, which allows each car and RSU to find out all neighboring cars and RSUs in real time. These information are broadcasted as periodic messages, which are also known as beacons. The beacons are generated and broadcasted consecutively to tell the neighbors about the car and RSU profile. Beacons can include these information about cars: speed, coordinate, next coordinate, acceleration, etc. The IEEE 802.11p protocol is used for communications in the VANET. IEEE 802.11p uses a Media Access Control (MAC) protocol that is based on a Carrier Sense Multiple Access protocol with Collision Avoidance

(CSM-A/CA) [25]. Beacons are sent on a fixed interval in VANETs. Most vehicular applications use 10Hz for generation rate; also a beacon is about 400 bytes long (including security fields) [25]-[26].

C. Simulation Scenario for VANET

As mentioned before, this study selects the Veins framework for simulation of VANET. Veins framework integrates SUMO and OMNET++. At first traffic of the cars are generated in SUMO and then exported to the network simulator. OMNET++ considers all cars as nodes and simulates the scenario. If any change occurs in the network, Veins can change the cars scenario in SUMO. Veins has a drawback which is that only cars are considered in OMNET++. This paper has developed Veins framework by adding a new module in OMNET++ which considers the traffic lights as nodes as well.

By considering this new module and Veins, realistic traffic of Cologne (and also the mentioned emergency cars) are simulated as a VANET in OMNET++. Every node can communicate by using IEEE 802.11p standard. The main parameters which have been used in this study for the network simulation are summarized in Table II. Simulation is done as follows: All the cars and also every traffic light broadcast beacons periodically, if a traffic light receives a beacon from an emergency car, the traffic light status changes to green for the emergency vehicle and red for the other cars, After the emergency car passes the traffic lights (with the distance of 20 meters) the traffic status returns to their original state. The simulation is done for all 20 emergency cars and response times are obtained for them. Using the parameters of Table II, yields a transmission range about 300 meters.

D. Simulation Result

The simulation has been done and the response times for all the emergency cars have been obtained. Fig.7 shows the response times for 20 emergency cars which are obtained by using car to infrastructure communication to change the traffic lights status with OMNET++.

TABLE II. NETWORK SIMULATION PARAMETERS

Beaconing Rate	10 Hz
Beacon size	400 Bytes
Maximum Transmission Power	20 mW
Thermal Noise	-110 dBm
Header Length	24bit
Slot Duration	16 µs
Beacon delivery deadline	100 ms
EIFS	188 µs
AIFS	64µs
CWmin	15
CWmax	1023
CCH interval duration	50 ms
Data rate	3 Mbit/s

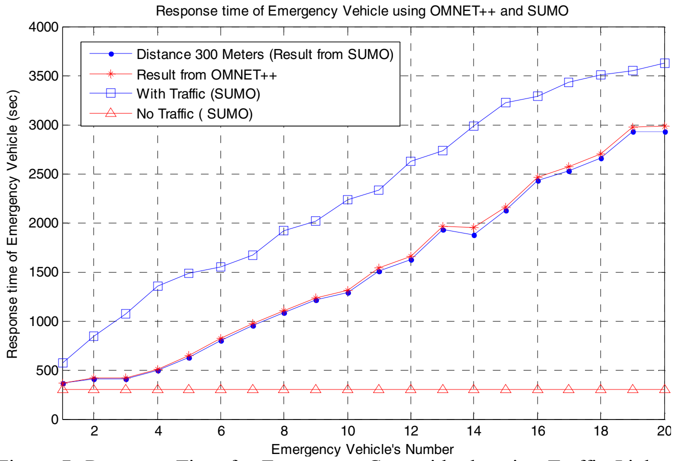

Figure 7. Response Time for Emergency Cars with changing Traffic Lights Status based on distance between cars and traffic lights in SUMO and using V2I communication to changing traffic lights status in OMNET++.

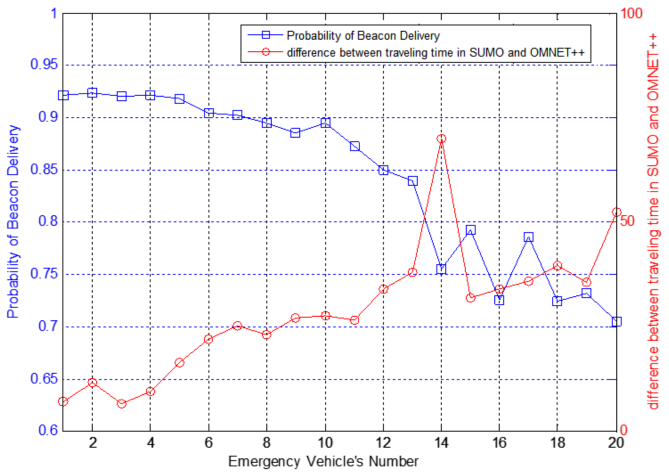

Figure 8. Probability of Beacon Delivery and the difference between response time in SUMO simulator (300 meters scenario) and OMNET++ simulator.

Also Fig.7 shows the data which were obtained in previous sections with using SUMO. Moreover, in order to analyze the accuracy of VANET simulation, Probability of Beacon Delivery for 20 emergency cars is calculated by considering all the received and lost beacons during the travel from origin to destination. Also for each emergency car the difference between response time in SUMO simulator (300 meters scenario) and OMNET++ simulator are considered. Figure.8 shows probability of beacon delivery for 20 emergency cars and also corresponding time differences.

Figure.7 and Fig.8, show this fact that the difference between VANET simulation result (realistic) and traffic simulation result (ideal) is quite small and almost the same. Realistic result is obtained from simulation in OMNET++ by using the beaconing and 802.11p standard and the traffic light status is changed based on the received beacons from emergency cars; and ideal form is obtained from SUMO by using the developed program which considers the exact distances between emergency cars and traffic lights for changing the traffic lights status. One of the reasons for this accurate result is related to beaconing scenario with numerous sending messages from each car which helps to deliver the message as soon as possible. This fact illustrates that car to car communication and car to infrastructure communication based on 802.11p is able to decrease the response time of the emergency cars. In addition Fig. 8 shows that by increasing the traffic density (emergency car's number) the probability of beacon delivery has decreased and also the emergency cars response time has increased.

V. Conclusion and Future work

In this paper, a new program has been developed which can be connected to SUMO and simulates a city with intelligent traffic lights that helps to decrease response time for emergency vehicles. This program can be used for any city and for any route for emergency vehicle to investigate effect of changing traffic lights status. Then, in order to obtain a realistic and reliable result, real map and realistic traffic of city of Cologne is selected and the developed program is simulated Cologne traffic with intelligent traffic lights. Response time for emergency cars in city of Cologne is investigated. Then this scenario is done by using network simulator OMNET++ and Veins framework. Probability of beacon delivery in real large area is obtained for emergency cars. VANET simulation and traffic simulation results have been compared. The results show that car to car communication can be implemented in real word by using 802.11p which helps to make roads safer and more comfortable.

Future work includes simulating traffic of the city with intelligent traffic lights with network simulators by using another wireless technologies such as WiMAX.

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